Inert Pyrolysis of Engineered Woods Lab Plan

Furnace & Atmosphere: Use a tube furnace capable of reaching ~500 °C, equipped with a continuous nitrogen purge. An inert nitrogen atmosphere (e.g. 1–2 L/min flow) is essential to prevent oxidation or combustion of the wood during heating. (In similar studies, wood was pyrolyzed under nitrogen at ~2 L/min to carry away gases.) The nitrogen purge also sweeps out volatile byproducts.

Sample Preparation: Each wood sample (provide ~10–100 g, depending on furnace size) should be placed in an open ceramic or stainless boat/crucible. Multiple trials – ideally triplicate runs for each wood type – will be done to ensure reproducibility.

Heating Program: After purging the system with N₂ (to displace air), ramp the temperature at a moderate rate (~10 °C/min) up to 500 °C. Once at 500 °C (± a chosen optimal setpoint), maintain that temperature for 30–60 minutes to ensure thorough pyrolysis. *Rationale:* Lower temperatures (~300 °C) yield higher char mass but indicate incomplete conversion (e.g. ~50% char yield at 300 °C means many volatiles remained). By 500 °C, most cellulose and hemicellulose have decomposed, and volatile tars and gases are driven off, leaving a carbon-rich char. (Note: extremely high temperatures >600 °C are not necessary for our purposes and could lead to excessive mineral volatilization.)

Residence & Cooling: Total residence time at 500 °C will be about 1 hour (including heating and hold). After the hold, cool the furnace under nitrogen flow until the sample is below ~100 °C. This prevents the hot char from oxidizing or igniting upon exposure to air. Only then can the nitrogen flow be stopped and the char sample retrieved.

Scrubber System for HCl Gas Removal

Why: During pyrolysis, any organically bound chlorine in the wood (e.g. from certain treatments or salts) will form hydrogen chloride gas (HCl) which must be removed for safety. HCl is highly corrosive and toxic, so we will scrub the exhaust gas to neutralize and capture it. The pyrolysis gas expulsion will also contain organic acids (like acetic acid from wood pyrolysis), which make the vapors acidic ("wood vinegar"). Removing these acidic gases protects the furnace and provides a cleaner char.

Scrubber Setup: Attach a gas outlet tube from the furnace to a gas-washing bottle (Drechsel bottle or impinger) containing an alkaline scrubbing solution. The recommended scrubbing medium is an aqueous sodium hydroxide (NaOH) solution at 2% (approximately 0.5 M). We choose 2% NaOH because NaOH readily neutralizes HCl (and other acids) into harmless salts ($HCl + NaOH \rightarrow NaCl + H_2O$). A dilute (0.5–3% NaOH) solution is typically used in industry for HCl scrubbing; ~1–2% provides a good buffering capacity while avoiding excessive causticity. At ~2% NaOH, the scrubber solution will stay in the optimal pH ~9–11 range to effectively capture HCl gas until fully spent.

Operation: As the furnace heats the wood, the evolving gases are carried by nitrogen through the NaOH scrubber. The gas should bubble through the alkaline solution, allowing HCl and acidic vapors to dissolve and neutralize. We will use at least one wash bottle (filled with ~200–300 mL of 2% NaOH); if available, a second in series acts as backup to ensure complete scrubbing. The scrubber should be vented (after the solution) into a fume hood or safe exhaust. Over the course of the pyrolysis, the NaOH will convert any HCl to sodium chloride in solution. This prevents HCl release into the lab and protects downstream equipment. (In a related setup, researchers used acid bubblers to capture inorganic contaminants from pyrolysis gas – here we instead use a base to specifically trap HCl.)

After the run, the scrubbing solution can be checked with pH paper; if it's still basic (pH > 7), it has neutralized the acids successfully. Expelling HCl is crucial because it prevents corrosion of metal parts and glassware and eliminates a hazard to personnel. The neutralization also ensures chlorine from the wood is mostly removed as salt solution, so the resulting char isn't chlorinated (beneficial for subsequent analysis).

Resulting Char Sample Characteristics

By the end of the pyrolysis, each wood sample will be converted to a dry, black carbonaceous char. This biochar will be brittle/friable and largely composed of carbon and ash (mineral content of the wood). We expect the char to retain inorganic compounds like metal oxides or salts from preservatives, now concentrated on its surface after organics are burned off. In fact, pyrolysis tends to precipitate and agglomerate metals and minerals within the char matrix. For example, one study on treated wood showed arsenic, copper, and chromium forming visible precipitates on the biochar surface. Visually, our char samples should be uniformly carbonized (no un-charred wood remaining), indicating complete pyrolysis. There should be no excess tar or moisture; all volatile resins, oils, and acids will have been either driven off or absorbed by the scrubber. The absence of HCl in the char (thanks to scrubbing) means the char is less corrosive and more stable for handling. We will store the char in sealed containers upon cooling (to avoid any oxidation from air exposure) until analysis.

Importance for XANES: These charred samples are ideal for XANES because they are solid, homogeneous, and enriched in the elements of interest. Removing the bulk of organic material (via pyrolysis) increases the relative concentration of target elements (e.g. metals like Cu, Cr, As from treatments, or Cl in salts) in the char, making them easier to detect spectroscopically. The char's carbon matrix is also relatively inert, which helps avoid interference during X-ray absorption measurements.

XANES Analysis Rationale (Brief)

X-ray Absorption Near-Edge Structure (XANES) spectroscopy will be used on the chars to determine the chemical speciation (oxidation state and bonding environment) of key elements (e.g. arsenic, chromium, copper, etc., depending on wood treatment). XANES is extremely valuable for this because it can distinguish, for instance, between As(V) and As(III) or Cr(VI) and Cr(III) in the sample. The oxidation state is crucial to know since toxicity and mobility of these elements depend on their chemical form (e.g. As(III) is more toxic/mobile than As(V)). We expect the pyrolysis process may alter some of these species – for example, prior studies found that arsenic pentavalent (As(V)) in CCA wood was partially reduced to As(III) in the char. By performing XANES on our chars, we can confirm such transformations and understand the state of the preservative elements after pyrolysis. This information is significant for environmental and analytical reasons: it tells us how stable or hazardous the remaining elements are in the char, and it demonstrates the efficacy of pyrolysis in changing or fixing those elements. In summary, XANES will provide insight into the chemical form of the elements concentrated in our biochar, helping interpret the impact of the pyrolysis treatment and guiding any further usage or disposal considerations for the char.

Sources:

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